

The COMS Randomized Trial of Iodine 125 Brachytherapy for Choroidal Melanoma

V. Twelve-Year Mortality Rates and Prognostic Factors: COMS Report No. 28

Collaborative Ocular Melanoma Study (COMS) Group*

Objectives: To report refined rates of death and related outcomes by treatment arm through 12 years after primary treatment of choroidal melanoma and to evaluate characteristics of patients and tumors as predictors of relative treatment effectiveness and time to death.

Design: Randomized multicenter clinical trial of iodine 125 (¹²⁵I) brachytherapy vs enucleation conducted as part of the Collaborative Ocular Melanoma Study. Eligible patients were free of metastasis and other cancers at enrollment. All patients were followed up for 5 to 15 years at scheduled examinations for metastasis or another cancer or until death. Decedents were classified by the independent Mortality Coding Committee as having histopathologically confirmed melanoma metastasis, suspected melanoma metastasis without histopathologic confirmation, another cancer but not melanoma metastasis, or no malignancy.

Main Outcome Measures: Deaths from all causes and deaths with histopathologically confirmed melanoma metastasis.

Results: Within 12 years after enrollment, 471 of 1317 patients died. Of 515 patients eligible for 12 years of follow-up, 231 (45%) were alive and clinically cancer free 12 years after treatment. For patients in both treatment

arms, 5- and 10-year all-cause mortality rates were 19% and 35%, respectively; by 12 years, cumulative all-cause mortality was 43% among patients in the ¹²⁵I brachytherapy arm and 41% among those in the enucleation arm. Five-, 10-, and 12-year rates of death with histopathologically confirmed melanoma metastasis were 10%, 18%, and 21%, respectively, in the ¹²⁵I brachytherapy arm and 11%, 17%, and 17%, respectively, in the enucleation arm. Older age and larger maximum basal tumor diameter were the primary predictors of time to death from all causes and death with melanoma metastasis.

Conclusion: Longer follow-up of patients confirmed the earlier report of no survival differences between patients whose tumors were treated with ¹²⁵I brachytherapy and those treated with enucleation.

Application to Clinical Practice: Estimated mortality rates by baseline characteristics should facilitate counseling of patients who have choroidal melanoma of a size and in a location suitable for enucleation or ¹²⁵I brachytherapy and no evidence of metastasis or another malignancy.

Trial Registration: clinicaltrials.gov Identifier: NCT00000124

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FROM 1986 TO 2003, THE COLLABORATIVE Ocular Melanoma Study (COMS) Group conducted 2 multicenter randomized clinical trials of radiotherapy vs enucleation for treatment of choroidal melanoma, a subset of uveal melanoma, the most common primary intraocular cancer in adults.¹ In the clinical trial that is the subject of this report, patients with unilateral choroidal melanoma who had tumors of a size and in a location such that eye-conserving brachytherapy was judged to be a treatment option were assigned randomly to irradiation of the tumor using iodine 125 (¹²⁵I) brachytherapy or to enucleation of

the eye.² When the COMS Group reported mortality findings from this trial through September 2000, the 5-year vital status was known for 1263 (96%) of the 1317 patients who enrolled.² Five-year all-cause mortality rates were 19% and 18% among patients assigned to enucleation and ¹²⁵I brachytherapy, respectively; 5-year rates of death with histopathologically confirmed melanoma metastasis were 11% and 9%, respectively. As reported at that time, the most important predictors of survival were younger age and smaller maximum basal tumor diameter (MBTD), consistent with earlier reports by others.³⁻⁶ Other outcomes and findings from this trial have been published.⁷⁻⁹ Most recently, find-

ings from a parallel study of quality of life, in which patients treated with enucleation and ^{125}I brachytherapy were compared on multiple instruments and scales, have been published.¹⁰ The companion COMS clinical trial, from which findings also have been reported,¹¹⁻¹⁴ was designed for patients who had unilateral choroidal melanoma and for whom most ophthalmologists would recommend enucleation owing to tumor size or location; these patients were assigned randomly to pre-enucleation radiation therapy or to enucleation alone.

The COMS clinical center personnel continued to follow up all patients in the ^{125}I brachytherapy trial until July 31, 2003. By that time, all patients had been followed up for mortality according to a standard protocol for 5 to 15 years. The purposes of this report are (1) to refine estimated rates of death and related events through 12 years of clinical follow-up, (2) to compare mortality rates between treatment arms within subgroups of patients defined by characteristics of the patients and ocular tumors documented at the time of the baseline examination, and (3) to provide precise estimates of mortality rates within those subgroups. This information may be useful to physicians and patients when considering treatment options for choroidal melanoma amenable to brachytherapy as used in the COMS and when projecting the number of years of remaining lifetime that a patient can expect statistically based on individual characteristics at the time of diagnosis and on the choice of primary treatment.

METHODS

The COMS design and many of the methods have been published.⁷⁻²⁰ Study documents with the details of procedures and data collected are available.^{21,22} Before enrollment and random assignment to treatment, all patients gave written consent to randomization and follow-up in the COMS for 10 years.

PATIENT ELIGIBILITY AND ENROLLMENT

Detailed eligibility and exclusion criteria and the methods used to evaluate patients for eligibility for the COMS trial of ^{125}I brachytherapy vs enucleation are described elsewhere.^{2,9,15} Eligible patients had unilateral choroidal melanoma with an apical height of 2.5 to 10.0 mm and an MBTD of 16.0 mm or less. Patients whose tumors were contiguous with the optic disc were ineligible. Patients for whom metastatic melanoma or other cancer, apart from nonmelanotic, noninvasive skin cancer or cancer in situ of the uterine cervix, could not be ruled out also were ineligible for the trial. Random assignment to ^{125}I brachytherapy or enucleation was made by designated personnel at the COMS Coordinating Center after a computer-assisted review of eligibility and telephone confirmation that the consent form had been signed. The details of the enrollment procedures, treatment protocols, and adherence are described elsewhere.²

DATA COLLECTION AND PATIENT FOLLOW-UP

Clinical examinations for data collection purposes were scheduled 6 and 12 months after enrollment, at 6-month intervals thereafter until completion of the 5-year examination, and then at 12-month intervals with optional examinations at the midpoint between annual examinations. During the 10th year of clinical follow-up of individual patients, each patient was given the op-

tion of continuing follow-up examinations by COMS clinical center personnel or of ceasing clinical follow-up on completion of the 10-year examination, with the further option in the latter case of follow-up for vital status using national databases or withdrawing from all COMS follow-up for clinical and vital status. Otherwise, patients were examined in accord with the COMS schedule until death or completion of the 15-year examination or the annual examination scheduled between August 1, 2002, and July 31, 2003, which was designated the final examination for COMS clinical data collection purposes. Patients were contacted by telephone at the midpoint between annual examinations whenever a clinical examination was not completed during that interval. Diagnoses of metastatic melanoma and of new primary cancers were reported as COMS clinical personnel made the diagnoses or received the information from non-COMS physicians.

Deaths, including those that occurred between the final scheduled study examinations of individual patients and July 31, 2003, were reported promptly to the COMS Coordinating Center as the personnel at the participating clinical centers became aware of them. In May 2005, a search of the National Death Index database (National Center for Health Statistics, US Department of Health and Human Services, Rockville, Md) was undertaken to confirm the vital status as of July 31, 2003, for all patients residing in the United States who had consented to transmittal of personal identifying information required for such searches to the COMS Coordinating Center.

Records for patients who were reported by COMS clinical personnel to have died by July 31, 2003, were reviewed by the COMS Mortality Coding Committee to classify status at death with respect to melanoma metastasis or another cancer.²⁰ Classification of decedents as having histopathologically confirmed melanoma metastasis was based on microscopic review of tissue, centrally at the COMS Pathology Center (Madison, Wis) or by the local pathologist. At their final meeting on January 16, 2004, the Mortality Coding Committee reviewed records of all patients diagnosed as having melanoma metastasis who were alive as of July 31, 2003.

To investigate clinically apparent disease progression over time, the status of each patient at the end of each year of follow-up, based on time from enrollment, was categorized on the basis of diagnoses reported by clinical personnel. The states considered included alive without a diagnosis of melanoma metastasis or another cancer; alive with a diagnosis of another cancer but not melanoma metastasis; alive with a diagnosis of melanoma metastasis; dead with no earlier diagnosis of melanoma metastasis or another cancer; dead after a diagnosis of another cancer but not melanoma metastasis; and dead after a diagnosis of melanoma metastasis. The state categorization of each deceased patient derived from clinical reports was compared with the status at death as judged by the Mortality Coding Committee. Because the database of deaths in the United States in 2003 assembled by the National Death Index was not available for searches until May 2005, more than 1 year after the final meeting of the Mortality Coding Committee and the end of COMS operations at participating clinical centers, clinical and other records of decedents identified only through the May 2005 search were not reviewed by the Mortality Coding Committee.

DATA ANALYSIS

We analyzed all deaths that occurred by July 31, 2003, and no more than 12 years after enrollment of individual patients. However, because of the smaller numbers of patients followed up for longer than 10 years, some analyses are limited to findings within 10 years of enrollment of individual patients. Data for each patient were analyzed with the treatment arm to which he or she was assigned randomly at the time of enrollment.

Unadjusted cumulative mortality rates (times to death) were calculated using the product-limit method²³; the log-rank test^{24,25} was used to compare treatment arms. Patients were censored from analysis after the last reported COMS examination or contact. To compare findings between treatment arms and to adjust mortality rates 5 and 10 years after treatment for important baseline covariates, we evaluated potential predictors of time to death by creating patient subgroups based on data collected before treatment was assigned. Of interest were predictors confirmed in earlier analysis of mortality in this trial² (patient age and MBTD), additional prognostic variables identified by other investigators,^{3,6} and other baseline characteristics of tumors investigated in the COMS randomized trial of pre-enucleation radiation.¹⁴ Exploration of 5- and 10-year mortality from all causes and death with melanoma metastasis overall and in patient subgroups resulted in 132 comparisons of mortality rates by treatment arm.

Subgroups based on patient characteristics measured on a continuous or an ordinal scale were defined using the median value or category to yield subgroups of similar size for data display purposes. Mortality rates within subgroups defined by dichotomous variables that contained fewer than 50 patients per treatment arm at entry were not estimated because of inherent instability of rates in small samples. The Cox proportional hazards model²⁶ was used to confirm that the patient's age and MBTD persisted as important predictors of time to death and to evaluate the importance of other potential baseline predictors. The proportional hazards model also was used to estimate hazard ratios (risk ratios) and 95% confidence intervals (CIs), to adjust mortality rates for covariates for comparison of treatment arms for all patients and within subgroups, and to evaluate potential covariate-treatment arm interactions. To estimate the risk ratio (risk of death in the ¹²⁵I brachytherapy arm vs the enucleation arm) and its 95% CI for death during the first 5 or 10 years after enrollment and treatment, patients were censored at 5 or 10 years after enrollment. Age and MBTD were used as continuous variables in all regression models.

No formal adjustment was made for the many comparisons of treatment arms; however, only probabilities less than .01 were deemed of potential interest (statistically significant) a priori. We used commercially available software (SAS, version 8.2; SAS, Inc, Cary, NC) for most data analyses and supplemented it with custom-written software.

RESULTS

As of July 31, 2003, all 1317 patients enrolled in this trial had been followed up for mortality for at least 5 years. Within the first 5 years, 252 patients died, including 127 (19%) of the 660 patients in the enucleation arm and 125 (19%) of the 657 patients in the ¹²⁵I brachytherapy arm. The estimated 95% CIs on the 5-year mortality rates were 16% to 22% in both treatment arms. The vital status 5 years after enrollment could not be ascertained for 4 patients, all in the enucleation arm; these 4 patients have been assumed to be alive in all analyses. One patient in the ¹²⁵I brachytherapy arm whose vital status at 5 years was unknown at the time of the initial publication of mortality findings² was ascertained to be living 8 years after enrollment. Of 799 patients who enrolled early enough to be eligible for 10 years of follow-up, the vital status at 10 years was unknown for 1 patient in the ¹²⁵I brachytherapy arm and 6 patients in the enucleation arm; these 7 patients also have been assumed to be alive.

TIME TO DEATH

Cumulative rates of death by time since enrollment are shown by treatment arm through 12 years in **Figure 1**. Cumulative all-cause mortality rates (Figure 1A) were similar in both treatment arms ($P = .70$, log-rank test), resulting in a pooled 5-year rate of 19% (95% CI, 17%-21%) and a 10-year rate of 35% (95% CI, 32%-38%). The cumulative 12-year all-cause mortality rate was less than 50% in each treatment arm (Figure 1A). The (unadjusted) risk ratios for ¹²⁵I brachytherapy vs enucleation were 1.00 (95% CI, 0.78-1.28) through the first 5 years after treatment, 1.01 (95% CI, 0.83-1.22) through 10 years after treatment, and 1.04 (95% CI, 0.86-1.24) during the entire 12-year period.

Of the 252 patients who died during the first 5 years of follow-up, 128 (10% of all patients and 51% of decedents) died with histopathologically confirmed melanoma metastasis diagnosed before or at the time of death. Of these deaths, 68 occurred among patients assigned to enucleation and 60 occurred among patients assigned to ¹²⁵I brachytherapy. Cumulative rates of death with histopathologically confirmed melanoma metastasis during 12 years after enrollment (Figure 1B) did not differ by treatment arm ($P = .62$, log-rank test), resulting in a pooled 5-year rate of 10% (95% CI, 9%-12%) and a 10-year rate of 17% (95% CI, 15%-20%). The unadjusted risk ratios were 0.89 (95% CI, 0.63-1.27) during the first 5 years, 1.01 (95% CI, 0.76-1.34) during 10 years, and 1.07 (95% CI, 0.81-1.41) through 12 years.

Combining deaths with histopathologically confirmed melanoma metastasis and deaths with suspected melanoma metastasis, 166 patients died with confirmed or suspected melanoma metastasis in the first 5 years after treatment. Five-year cumulative mortality rates with confirmed or suspected melanoma metastasis were 13% in both the enucleation and ¹²⁵I brachytherapy arms; 10-year rates were 21% and 22%, respectively.

PREDICTORS OF TIME TO DEATH

The only predictors of time to death among those evaluated, including the treatment arm, that met our criterion for statistical significance were patient age at baseline and MBTD. No interaction was found between these variables and the treatment arm. In **Figure 2**, cumulative mortality rates are given for the 4 subgroups of patients defined by age (≤ 60 and > 60 years) and MBTD (≤ 11 and > 11 mm) with both treatment arms combined. Older age conferred greater risk of death from all causes for smaller and larger tumors (Figure 2A). Cumulative rates of death from all causes by time since enrollment and treatment arm after adjustment for age and MBTD were nearly identical to the unadjusted mortality rates shown in Figure 1A. After adjustment, the risk ratio for death from all causes, for ¹²⁵I brachytherapy vs enucleation, was 1.03 (95% CI, 0.81-1.32) through 5 years and 1.03 (95% CI, 0.85-1.25) through 10 years, confirming no overall difference in all-cause mortality rates by treatment arm.

Patients whose MBTD was no larger than 11.0 mm, regardless of age, had the lowest 10-year rates of death

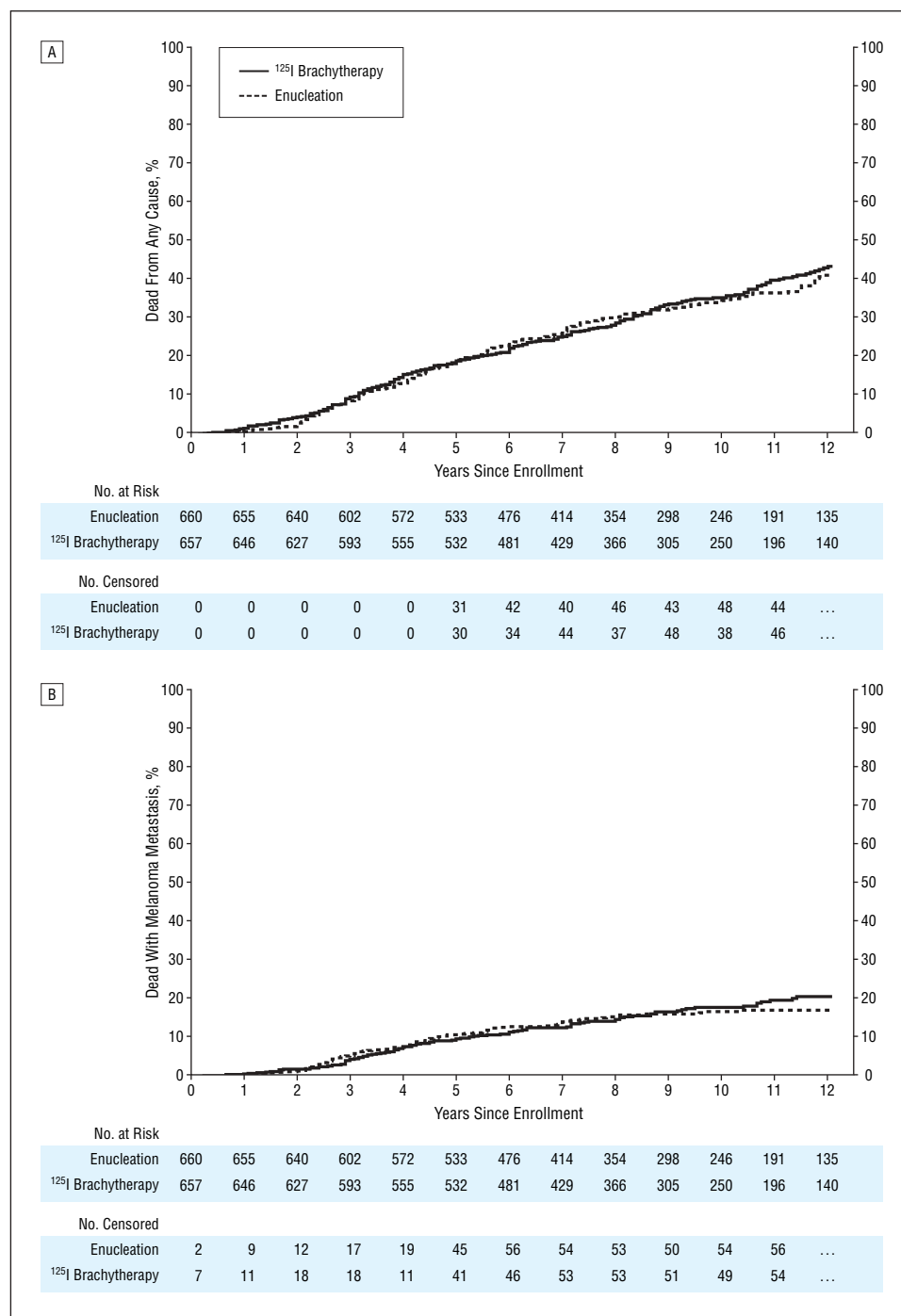
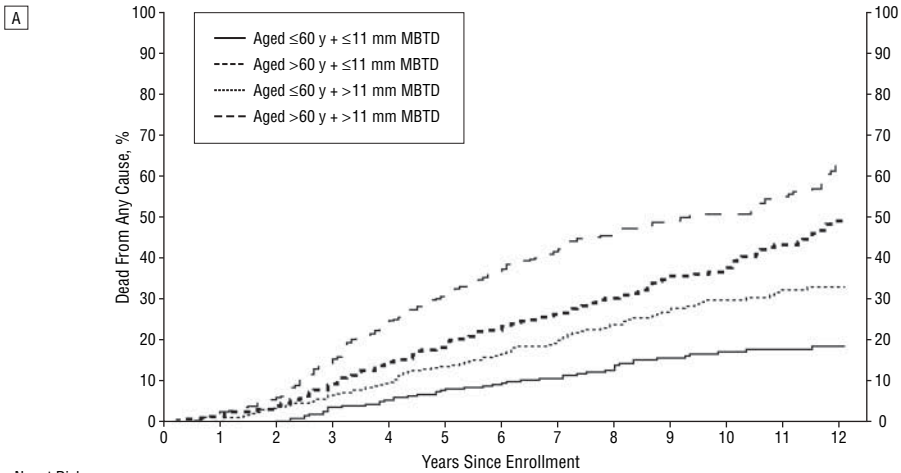


Figure 1. Cumulative percentage of patients dead by time since enrollment and treatment arm (iodine 125 [¹²⁵I] brachytherapy [n=657] and enucleation [n=660]). The numbers of patients at risk and censored at the end of each 1-year interval are shown below the horizontal axis. Ellipses indicate not applicable. A, Deaths from all causes. B, Deaths with histopathologically confirmed melanoma metastasis.

with histopathologically confirmed melanoma metastasis (10%; 95% CI, 7%-14%) (Figure 2B). Cumulative rates of death with melanoma metastasis by treatment arm after adjustment for age and MBTD were indistinguishable from the unadjusted rates shown in Figure 1B. After adjustment for age and MBTD, the risk ratio for death with histopathologically confirmed melanoma metastasis for ¹²⁵I brachytherapy vs enucleation was 0.91 (95% CI, 0.64-1.29) during the first 5 years after treatment and 1.01 (95% CI, 0.77-1.34) during 10 years, again suggesting no clinically or statistically important difference between treatment arms.

COMPARISON OF MORTALITY RATES BY TREATMENT ARM WITHIN PATIENT SUBGROUPS

Mortality rates by treatment arm were compared within subgroups defined by patient and tumor characteristics. Observed 5- and 10-year mortality rates within the 2 age subgroups were adjusted for MBTD; rates in the 2 subgroups defined by MBTD were adjusted for patient age. Mortality rates for the total group of patients and for each of the other patient subgroups were adjusted for both patient age and MBTD. Adjusted risk ratios for death in the



No. at Risk

MBTD > 11 mm

Age > 60 y	374	364	351	316	281	256	213	180	153	130	108	75	44
Age ≤ 60 y	314	313	302	293	283	271	249	218	186	148	127	104	79

MBTD ≤ 11 mm

Age > 60 y	338	333	324	306	288	271	245	216	179	144	115	88	57
Age ≤ 60 y	291	291	290	280	275	267	250	229	202	181	146	120	95

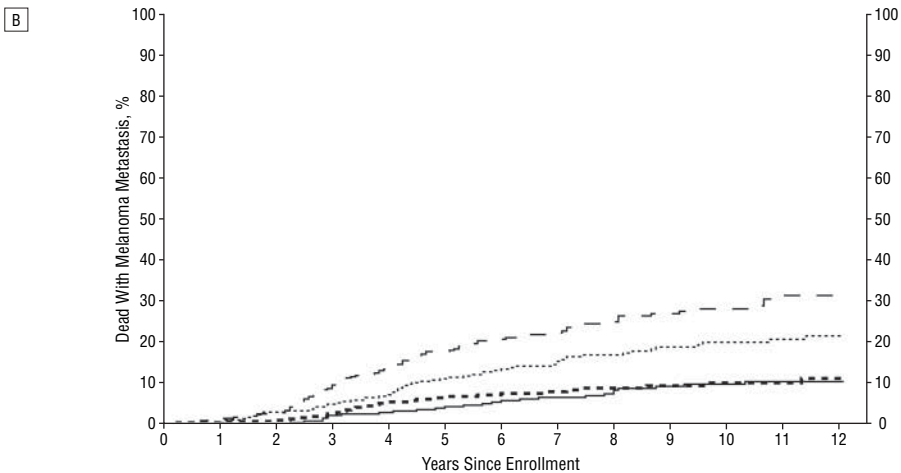
No. Censored

MBTD > 11 mm

Age > 60 y	0	0	0	0	0	22	17	16	15	17	25	19	...
Age ≤ 60 y	0	0	0	0	0	12	22	22	29	17	19	24	...

MBTD ≤ 11 mm

Age > 60 y	0	0	0	0	0	14	19	27	22	25	17	23	...
Age ≤ 60 y	0	0	0	0	0	13	18	19	17	32	25	24	...



No. at Risk

MBTD > 11 mm

Age > 60 y	374	364	351	316	281	256	213	180	153	130	108	75	44
Age ≤ 60 y	314	313	302	293	283	271	249	218	186	148	127	104	79

MBTD ≤ 11 mm

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Age ≤ 60 y	291	291	290	280	275	267	250	229	202	181	146	120	95

No. Censored

MBTD > 11 mm

Age > 60 y	5	9	10	20	13	34	29	21	19	20	29	31	...
Age ≤ 60 y	0	3	3	3	0	15	26	28	34	19	22	24	...

MBTD ≤ 11 mm

Age > 60 y	4	7	13	9	13	24	28	35	34	28	27	30	...
Age ≤ 60 y	0	1	4	3	4	13	19	23	19	34	25	25	...

Figure 2. Cumulative percentage of patients dead by time since enrollment and patient age and maximum basal tumor diameter (MBTD). Treatment arms were combined to estimate cumulative percentages. The numbers of patients at risk and censored at the end of each 1-year interval are shown below the horizontal axis. Ellipses indicate not applicable. A, Deaths from all causes. B, Deaths with histopathologically confirmed melanoma metastasis.

¹²⁵I brachytherapy arm vs the enucleation arm and their 95% CIs for the first 5 years of follow-up and for all 10 years are displayed by patient subgroup in eFigure 1 (available at <http://www.archophthalmol.com>) for deaths from all causes and in eFigure 2 (available at <http://www.archophthalmol.com>) for deaths with histopathologically confirmed melanoma metastasis. For all treatment arm comparisons within the patient subgroups examined, except those based on tumor shape and history of cardiovascular or circulatory disease at baseline, the 95% CIs for the risk ratio include 1.00, indicating that the underlying risk of death during each period was similar in both treatment arms. The interactions between treatment and tumor shape and between treatment and history of cardiovascular or circulatory disease met the conventional criterion for statistical significance ($P < .05$) but not our more stringent criterion. Examination of distributions of these 2 characteristics between treatment arms within the subgroups defined by age and MBTD at baseline revealed chance imbalances, with more patients who had mushroom-shaped tumors and more patients who had a history of cardiovascular or circulatory disease in the age-MBTD subgroup with the lowest mortality rates assigned to enucleation ($P = .07$, χ^2 test for homogeneity, for both comparisons). Adjustment of 5- and 10-year mortality rates for these 2 baseline covariates, in addition to age and MBTD, resulted in no change or trivial changes in risk ratios and CIs in other patient subgroups.

MORTALITY RATES BY PATIENT CHARACTERISTICS

Because differences in mortality rates between treatment arms were found only when patients were grouped by baseline tumor shape or by history of cardiovascular or circulatory disease, and because those differences did not meet our criterion for statistical significance, patients in both treatment arms were combined to provide estimates of mortality rates that were as precise as possible within subgroups, after adjustment for patient age and MBTD. **Figure 3** summarizes adjusted estimates of cumulative mortality rates for individual patient subgroups for deaths from all causes; **Figure 4** provides similar information for deaths with histopathologically confirmed melanoma metastasis. The vertical lines and shaded bars in these displays indicate the adjusted cumulative mortality rates at 5 and 10 years and their 95% CIs for the total group of 1317 patients. Subgroups for which the CIs lie to the right of the CI for the total group are those in whom mortality rates were somewhat higher than for all of the patients enrolled. Subgroups for whom the CIs lie to the left of the CI for the total group are those for whom mortality rates were somewhat lower than for all of the patients enrolled.

Comparison of Figures 3 and 4 indicates that the difference in all-cause mortality rates between the subgroups of patients older than 60 years and those 60 years or younger at baseline is owing to deaths without histopathologically confirmed melanoma metastasis. This observation was investigated by a detailed examination of classifications by the Mortality Coding Committee of decedents by age at baseline. The proportion of deaths with

confirmed melanoma metastasis was lower in the older baseline subgroup at enrollment, but the higher number of deaths from all causes among older patients resulted in similar rates of death with histopathologically confirmed melanoma metastasis in both age subgroups. A similar observation, investigation, and conclusion apply to patients with and without a history of cardiovascular or circulatory disease at baseline.

VITAL STATUS BY DIAGNOSIS OF MELANOMA METASTASIS AND OTHER CANCERS

Figure 5 displays the status of patients at the end of each year of follow-up for both treatment arms combined. The number of patients followed up to the specified time is shown at the top of each bar. By 5 years after enrollment, 189 (14%) of the 1317 patients had been diagnosed as having melanoma metastasis; 159 of these patients had died, whereas the other 30 were alive. Of the 799 patients followed up for 10 years, 182 (23%) had been diagnosed as having melanoma metastasis, and 169 of these patients had died. Diagnosis of another primary cancer without a diagnosis of melanoma metastasis was less common; 109 patients (8%) had been diagnosed as having another cancer but not melanoma metastasis during 5 years after enrollment. Of the 515 patients followed up for 12 years, 285 (55%) were alive, and 231 (45%) were alive and free of clinically detectable cancer.

Classifications of decedents by the Mortality Coding Committee were compared with the clinical diagnoses of melanoma metastasis or other cancers before death (eTable, available at <http://www.archophthalmol.com>). Of the 251 decedents who had been diagnosed as having metastatic melanoma before death, the Mortality Coding Committee concluded that 201 (80%) had died with histopathologically confirmed melanoma metastasis. Of the remaining patients diagnosed as having melanoma metastasis while alive, 46 (18%) were designated as dying with a malignant tumor but without histopathologic confirmation of melanoma metastasis provided to the committee, that is, with suspected melanoma metastasis. Of 40 decedents who had been clinically diagnosed as having another primary cancer but never melanoma metastasis during clinical follow-up, 17 (42%) were classified as having a malignant tumor other than metastatic melanoma at death, and 3 (8%) were classified by the Mortality Coding Committee as having suspected (not histopathologically confirmed) melanoma metastasis at death. Of the 180 decedents who were not clinically diagnosed as having melanoma metastasis or another primary cancer, 35 (19%) were judged by the Mortality Coding Committee to have died with undiagnosed metastatic melanoma or another malignant tumor.

COMMENT

Comparisons of ¹²⁵I brachytherapy with enucleation to treat choroidal melanoma identified no clinically or statistically meaningful differences in mortality rates between treatment arms for up to 12 years after treatment, regardless of whether deaths from all causes (Figure 1A)

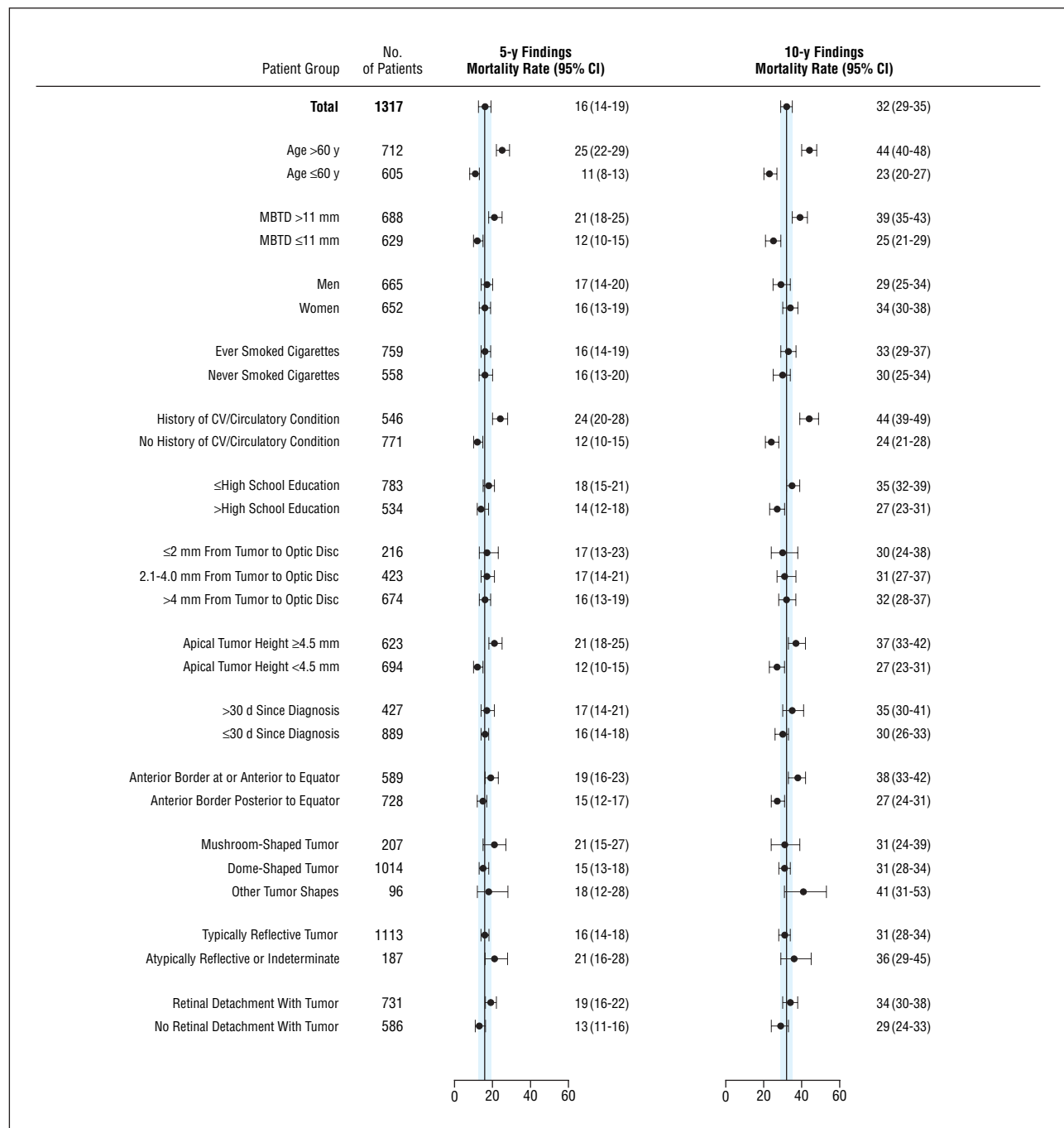


Figure 3. Adjusted cumulative rates of death from all causes by 5 and 10 years after enrollment within patient subgroups defined by baseline characteristics. Treatment arms were combined to estimate adjusted mortality rates. Rates in subgroups defined by patient age at baseline were adjusted for maximum basal tumor diameter (MBTD); rates in subgroups defined by MBTD were adjusted for patient age. Rates in all other subgroups were adjusted for patient age and MBTD. The vertical line shown for each interval indicates the adjusted cumulative all-cause mortality rate for the total group of 1317 patients; the shaded bar shows the 95% confidence interval (CI) associated with that rate. CV indicates cardiovascular. Information required to classify patients was not available for distance from the proximal tumor border to the optic disc (n = 4), days from diagnosis to enrollment (n = 1), and tumor reflectivity (n = 7).

or deaths with histopathologically confirmed melanoma metastasis (Figure 1B) were considered. Adjustment for important predictors of time to death, ie, age at time of enrollment and MBTD, did not alter this conclusion. These findings are from the largest study and, to our knowledge, the only randomized trial conducted to compare ¹²⁵I brachytherapy with enucleation with respect to survival. Perhaps the most important finding from this study is that nearly half (45%) of patients with cho-

roidal melanoma who met eligibility criteria for this trial, who were treated with enucleation or with ¹²⁵I brachytherapy, and who enrolled early enough to be eligible for 12 years of follow-up were alive and had remained clinically cancer free 12 years later (Figure 5).

The large number of patients enrolled and followed up in this clinical trial permitted precise estimation of mortality rates by time after treatment. When findings from patients in both treatment arms were combined, 95% CIs for

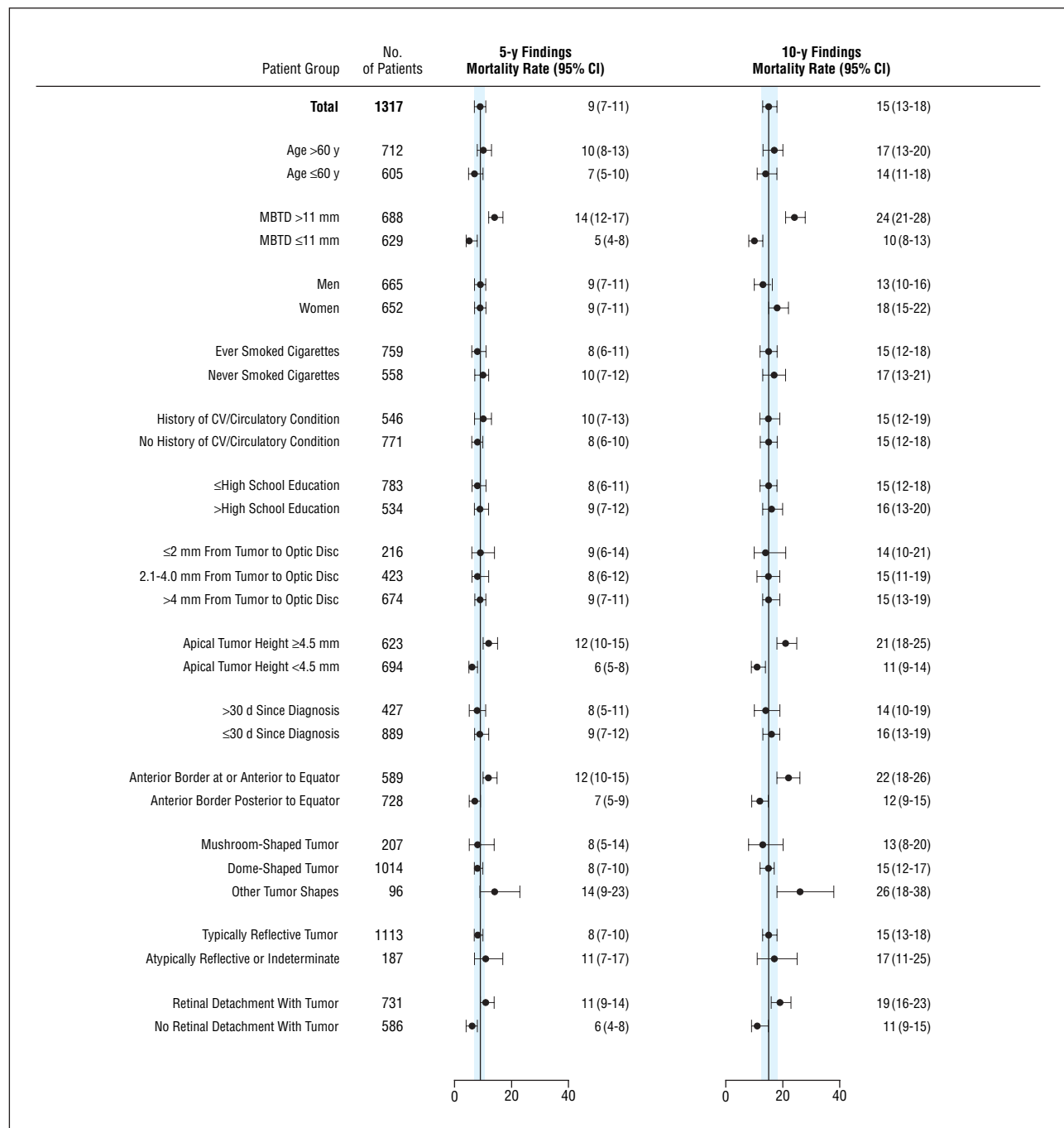


Figure 4. Adjusted cumulative rates of death with histopathologically confirmed melanoma metastasis by 5 and 10 years after enrollment within patient subgroups defined by baseline characteristics. Treatment arms were combined to estimate adjusted mortality rates. Rates in subgroups defined by patient age at baseline were adjusted for maximum basal tumor diameter (MBTD); rates in subgroups defined by MBTD were adjusted for patient age. Rates in all other subgroups were adjusted for patient age and MBTD. The vertical line for each interval indicates the adjusted cumulative rate of death with histopathologically confirmed melanoma metastasis for the total group of 1317 patients; the shaded bar shows the 95% confidence interval (CI) associated with that rate. CV indicates cardiovascular. Information required to classify patients was not available for distance from the proximal tumor border to the optic disc (n = 4), days from diagnosis to enrollment (n = 1), and tumor reflectivity (n = 7).

5- and 10-year mortality rates by subgroup typically were only 6% to 8% in width (Figures 3 and 4). Because patient age and MBTD were the most predictive baseline variables, mortality rates for subgroups defined by these characteristics should be those most useful to physicians when counseling patients who already have choroidal melanoma of similar size and location at the time of diagnosis and who satisfy other eligibility criteria for this trial. Sub-

group findings based on other characteristics of the patient and tumor should be interpreted cautiously to avoid conveying unwarranted optimism (or pessimism) regarding the number of years of life that a patient with those characteristics can expect.

Because only 26 patients (2%) reported their race or ethnicity to be other than non-Hispanic white, findings should be interpreted as applying primarily to non-

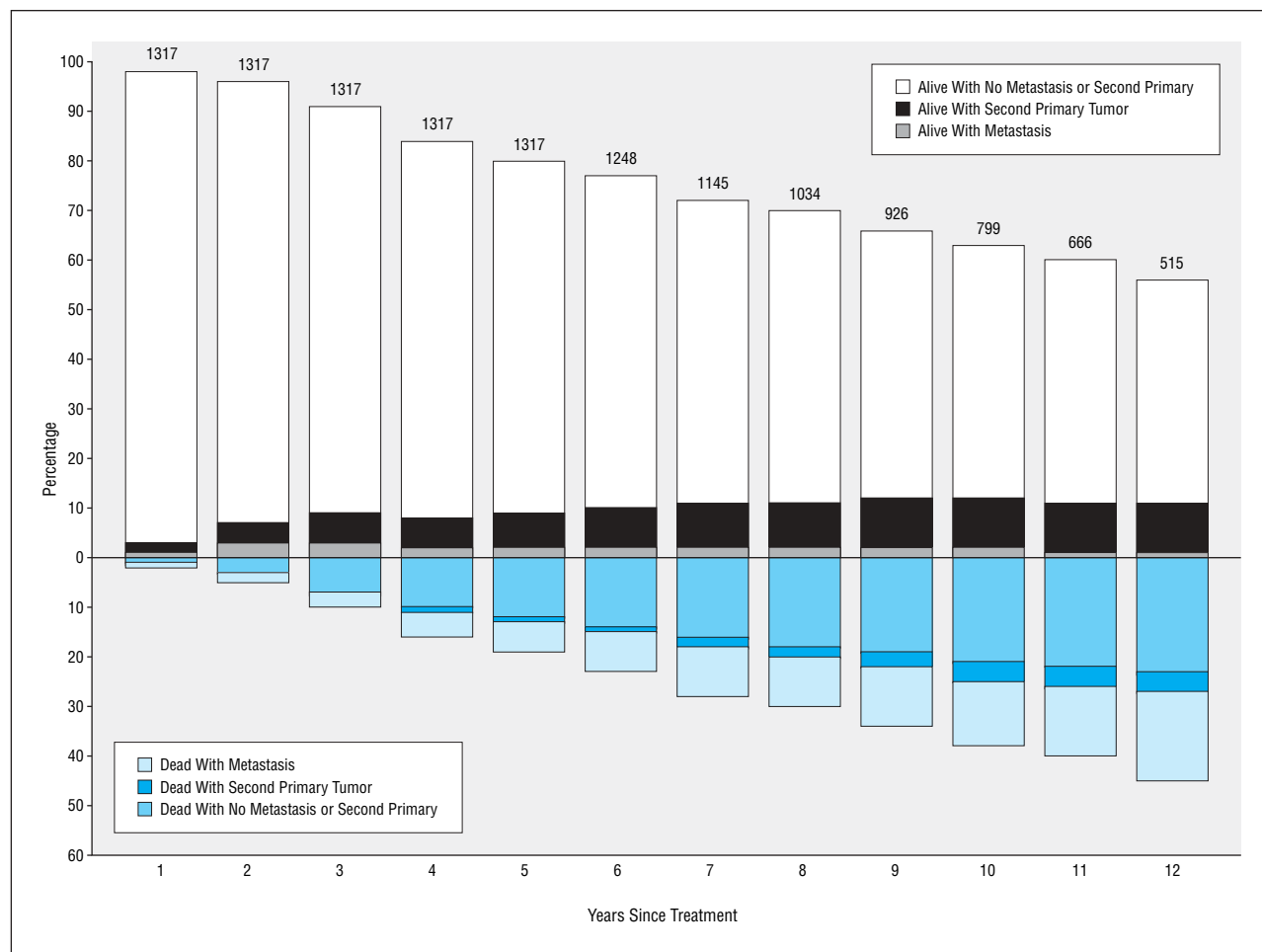


Figure 5. Percentage of patients with specified status at the end of each year of follow-up. Treatment arms were combined. The number of patients who enrolled early enough to be followed up to the end of the specified year is shown at the top of the corresponding bar.

Hispanic white patients. Similarly, because the posterior border of the tumor was anterior to the equator in only 38 patients, and because the anterior border was located posterior to the equator in 728 patients (55%), these findings apply largely to patients who have posterior choroidal melanoma.

This COMS clinical trial had greater than 80% power to detect differences in 5-year all-cause mortality rates between treatment arms such as 20% vs 25% or 15% vs 20% and nearly 70% power to detect similar differences within most subgroups. Thus, a difference of this magnitude is unlikely. Post hoc calculations of power also indicate that, with 799 patients followed up for mortality for 10 years, this clinical trial had better than 90% power to detect differences such as 30% vs 40% or 35% vs 45% and at least 80% power to detect differences such as these in most subgroups.

At the time this randomized trial was designed, a review of the published data regarding survival after enucleation of eyes with choroidal melanoma of medium size suggested that the 5-year mortality rate among patients assigned to the enucleation arm would be about 30%.²⁷ However, the COMS Group has shown that the prognosis after primary treatment with ¹²⁵I brachytherapy or enucleation for most patients diagnosed as having choroidal melanoma who meet the eligibility criteria adopted

by the COMS is much better than that prediction. Only patients in the highest-risk subgroup (aged >60 years with MBTD >11 mm) (Figure 2A) had a 5-year all-cause mortality rate close to 30%. After adjustment for age and MBTD, no other patient subgroup had a 5-year mortality rate close to 30% or a rate for which the 95% CI included 30% (Figure 4). Indeed, patients in the 2 age-MBTD subgroups at lowest risk of death during 12 years of posttreatment follow-up had 10-year estimated all-cause mortality rates of 30% or less (Figure 2A).

Although the findings from this clinical trial may tempt physicians and patients to extrapolate them to larger choroidal melanoma and to other patients who would not have met COMS eligibility criteria, we do not recommend such extrapolation. As already reported, larger tumor size and tumor location near the optic disc were associated with greater loss of visual acuity and higher rates of local complications that resulted in failure to achieve the goal of useful visual acuity in eyes treated with ¹²⁵I brachytherapy.^{7,8} Other methods of delivering radiation to the tumor and different treatment approaches may result in lower complication rates. However, these methods typically have not been shown to result in survival rates similar to or better than those achieved with enucleation or ¹²⁵I brachytherapy by direct comparison in randomized clinical trials. As reported elsewhere, quality of

life, as measured by both generic and vision-targeted instruments, was similar among COMS patients in the enucleation and ¹²⁵I brachytherapy arms by 5 years after treatment.¹⁰ Thus, patients and physicians should consider the strength of available evidence regarding the prognosis for survival and vision, as well as individual goals and values when selecting primary treatment for choroidal melanoma.

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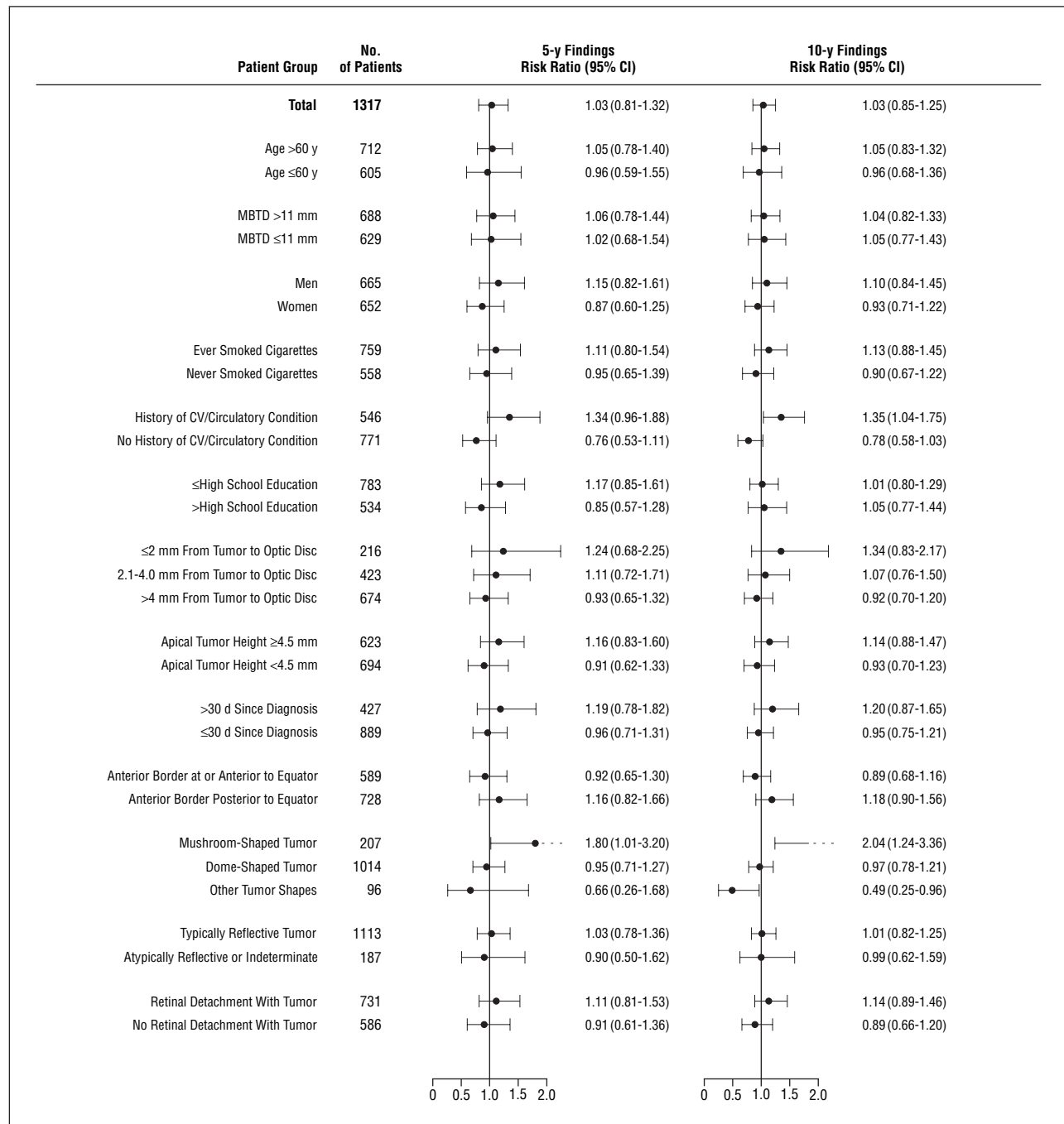
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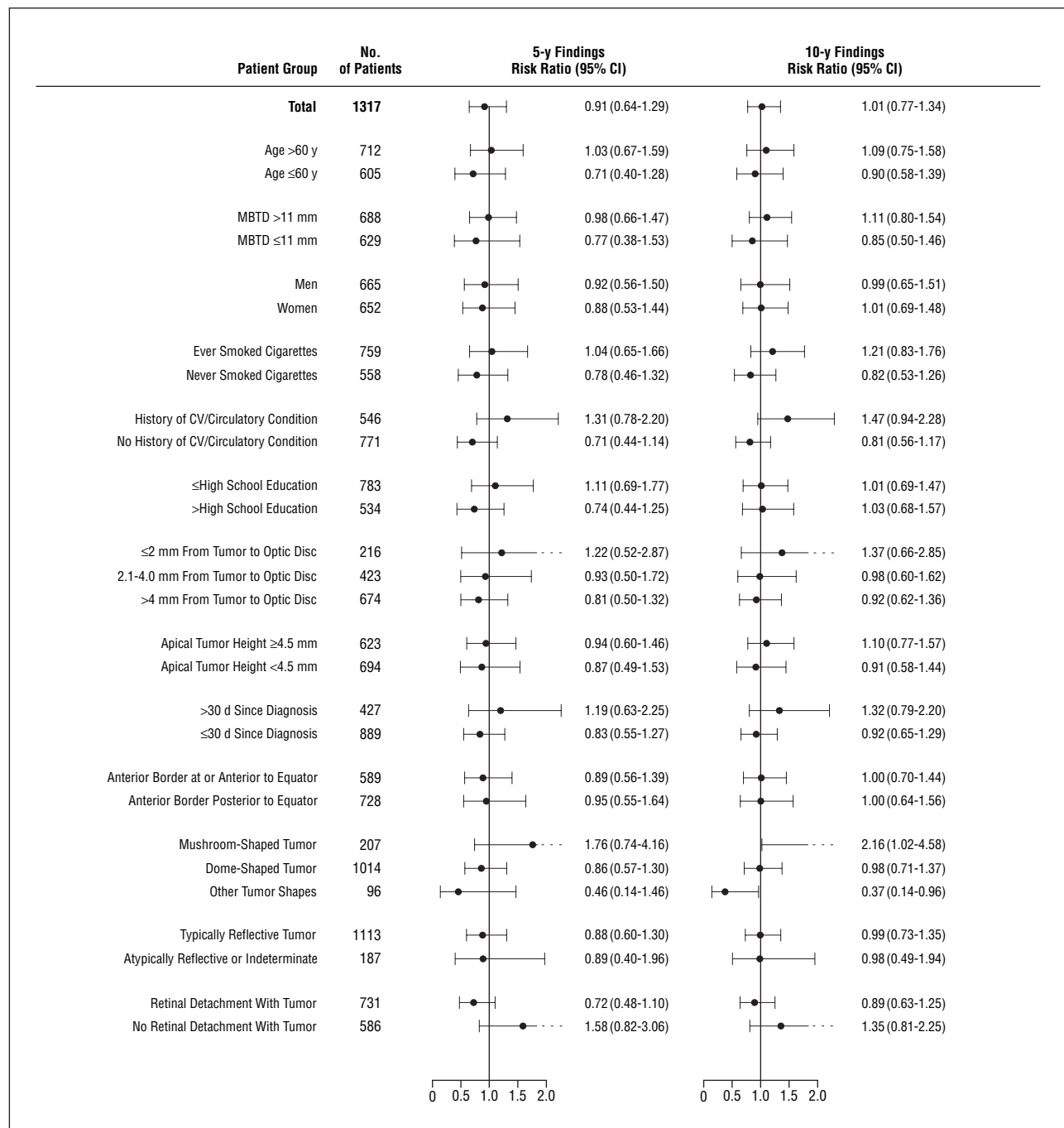
Additional Information: The online-only eFigures and eTable are available at <http://www.archophthalmol.com>.

REFERENCES

- Scotto J, Fraumeni JF, Lee JAH. Melanomas of the eye and other noncutaneous sites: epidemiologic aspects. *J Natl Cancer Inst*. 1976;56:489-491.
- Collaborative Ocular Melanoma Study Group. The COMS randomized trial of iodine 125 brachytherapy for choroidal melanoma, III: initial mortality findings: COMS report No. 18. *Arch Ophthalmol*. 2001;119:969-982.
- Shammas HF, Blodi FC. Prognostic factors in choroidal and ciliary body melanomas. *Arch Ophthalmol*. 1977;95:63-69.
- Seddon JM, Albert DM, Lavin PT, Robinson N. A prognostic factor study of disease-free interval and survival following enucleation for uveal melanoma. *Arch Ophthalmol*. 1983;101:1894-1899.
- Augsburger JJ, Gamel JW, Sardi VF, Greenberg RA, Shields JA, Brady LW. Enucleation vs cobalt plaque radiotherapy for malignant melanomas of the choroid and ciliary body. *Arch Ophthalmol*. 1986;104:655-661.
- Flocks M, Gerende JH, Zimmerman LE. The size and shape of malignant melanomas of the choroid and ciliary body in relation to prognosis and histologic characteristics: a statistical study of 210 tumors. *Trans Am Acad Ophthalmol Otolaryngol*. 1955;59:740-758.
- Collaborative Ocular Melanoma Study Group. Collaborative Ocular Melanoma Study (COMS) randomized trial of I-125 brachytherapy for medium choroidal melanoma, I: visual acuity after 3 years: COMS report No. 16. *Ophthalmology*. 2001;108:348-366.
- Collaborative Ocular Melanoma Study Group. The COMS randomized trial of iodine 125 brachytherapy for choroidal melanoma, IV: local treatment failure and enucleation in the first 5 years after brachytherapy: COMS report No. 19 [published correction appears in *Ophthalmology*. 2004;111:1514]. *Ophthalmology*. 2002;109:2197-2206.
- Collaborative Ocular Melanoma Study Group. The COMS randomized trial of iodine 125 brachytherapy for choroidal melanoma, II: characteristics of patients enrolled and not enrolled: COMS report No. 17. *Arch Ophthalmol*. 2001;119:951-965.
- Collaborative Ocular Melanoma Study Quality-of-Life Study (COMS-QOLS) Group. Quality of life after I-125 brachytherapy versus enucleation for choroidal melanoma: 5-year results from the Collaborative Ocular Melanoma Study: COMS QOLS report No. 3. *Arch Ophthalmol*. 2006;124:226-238.
- Collaborative Ocular Melanoma Study Group. The Collaborative Ocular Melanoma Study (COMS) randomized trial of pre-enucleation radiation of large choroidal melanoma, I: characteristics of patients enrolled and not enrolled: COMS report No. 9. *Am J Ophthalmol*. 1998;125:767-778.
- Collaborative Ocular Melanoma Study Group. The Collaborative Ocular Melanoma Study (COMS) randomized trial of pre-enucleation radiation of large choroidal melanoma, II: initial mortality findings: COMS report No. 10 [published correction appears in *Am J Ophthalmol*. 1998;126:622]. *Am J Ophthalmol*. 1998;125:779-796.
- Collaborative Ocular Melanoma Study Group. The Collaborative Ocular Melanoma Study (COMS) randomized trial of pre-enucleation radiation of large choroidal melanoma, III: local complications and observations following enucleation: COMS report No. 11. *Am J Ophthalmol*. 1998;126:362-372.
- Collaborative Ocular Melanoma Study Group. The Collaborative Ocular Melanoma Study (COMS) randomized trial of pre-enucleation radiation of large choroidal melanoma, IV: ten-year mortality findings and prognostic factors: COMS report No. 24. *Am J Ophthalmol*. 2004;138:936-951.
- Collaborative Ocular Melanoma Study Group. Design and methods of a clinical trial of a rare condition: the Collaborative Ocular Melanoma Study: COMS report No. 3. *Control Clin Trials*. 1993;14:362-391.
- Collaborative Ocular Melanoma Study Group. Histopathologic characteristics of uveal melanoma in eyes enucleated from the Collaborative Ocular Melanoma Study: COMS report No. 6. *Am J Ophthalmol*. 1998;125:745-766.
- Collaborative Ocular Melanoma Study Group. Echography (ultrasound) procedures for the Collaborative Ocular Melanoma Study (COMS): COMS report No. 12, part I. *J Ophthalmic Nurs Technol*. 1999;18:143-149.
- Collaborative Ocular Melanoma Study Group. Echography (ultrasound) procedures for the Collaborative Ocular Melanoma Study (COMS): COMS report No. 12, part II. *J Ophthalmic Nurs Technol*. 1999;18:219-232.
- Collaborative Ocular Melanoma Study Group. Consistency of observations from echograms made centrally in the Collaborative Ocular Melanoma Study: COMS report No. 13. *Ophthalmic Epidemiol*. 2002;9:11-27.
- Collaborative Ocular Melanoma Study Group. Cause-specific mortality coding: methods in the Collaborative Ocular Melanoma Study: COMS report No. 14. *Control Clin Trials*. 2001;22:248-262.
- Collaborative Ocular Melanoma Study Group. *COMS Manual of Procedures*. Springfield, Va: National Technical Information Service; January 1995. Accession No. PB95-179693.
- Collaborative Ocular Melanoma Study Group. *COMS Forms Book*. Springfield, Va: National Technical Information Service; August 1998. Accession No. PB98-139629.
- Kaplan EL, Meier P. Nonparametric estimation from incomplete observations. *J Am Stat Assoc*. 1958;53:457-481.
- Mantel N. Evaluation of survival data and two new rank order statistics arising in its consideration. *Cancer Chemother Rep*. 1966;50:163-170.
- Peto R, Peto J. Asymptotically efficient rank invariant procedures. *J R Stat Soc A*. 1972;135:185-206.
- Cox DR. Regression models and life-tables. *J R Stat Soc B*. 1972;34:187-220.
- Diener-West M, Hawkins BS, Markowitz JA, Schachat AP. A review of mortality from choroidal melanoma, II: a meta-analysis of 5-year mortality rates following enucleation, 1966 through 1988. *Arch Ophthalmol*. 1992;110:245-250.

eFigure 1, eFigure 2, and eTable.

eFigure 1. Adjusted risk ratios and 95% confidence intervals (CIs) for iodine 125 (¹²⁵I) brachytherapy vs enucleation for deaths from any cause during the first 5 years after enrollment and through 10 years within subgroups of patients defined by baseline characteristics. Mortality rates within subgroups based on patient age were adjusted for maximum basal tumor diameter (MBTD); rates within subgroups based on MBTD were adjusted for patient age. Mortality rates in all other subgroups were adjusted for age at enrollment and MBTD. The vertical line for each interval is located at 1.00, the value of the risk ratio (hazard ratio) that indicates no difference between treatment arms with respect to all-cause mortality. Risk ratios less than 1.00 indicate fewer deaths after ¹²⁵I brachytherapy, whereas risk ratios greater than 1.00 indicate more deaths after ¹²⁵I brachytherapy, in comparison with enucleation. The overall adjusted risk ratio for the total group of patients enrolled is shown first for comparison. CV indicates cardiovascular. Information required to classify patients was not available for distance from the proximal tumor border to the optic disc (n = 4), days from diagnosis to enrollment (n = 1), and tumor reflectivity (n = 7).



eFigure 2. Adjusted risk ratios and 95% confidence intervals (CIs) for iodine 125 (¹²⁵I) brachytherapy vs enucleation for deaths with histopathologically confirmed melanoma metastasis during the first 5 years after enrollment and through 10 years within subgroups of patients defined by baseline characteristics. Mortality rates within subgroups based on patient age were adjusted for maximum basal tumor diameter (MBTD); rates within subgroups based on MBTD were adjusted for patient age. Mortality rates in all other subgroups were adjusted for age at enrollment and MBTD. The vertical line for each interval is located at 1.00, the value of the risk ratio (hazard ratio) that indicates no difference between treatment arms with respect to death with histopathologically confirmed melanoma metastasis. Risk ratios less than 1.00 indicate fewer deaths after ¹²⁵I brachytherapy, whereas risk ratios greater than 1.00 indicate more deaths after ¹²⁵I brachytherapy, in comparison with enucleation. The overall adjusted risk ratio for the total group of patients enrolled is shown first for comparison. CV indicates cardiovascular. Information required to classify patients was not available for distance from the proximal tumor border to the optic disc (n = 4), days from diagnosis to enrollment (n = 1), and tumor reflectivity (n = 7).

eTable. Comparison of Classification of Decedents by the COMS Mortality Coding Committee With Diagnoses of Melanoma Metastasis or Other Cancer Before Death

Mortality Coding Committee Classification	No. (%) of Decedents With Specified Diagnosis Before Death		
	Melanoma Metastasis (n = 251)	Second Primary Cancer, Not Melanoma Metastasis (n = 40)	No Melanoma Metastasis or Other Cancer (n = 180)
Melanoma metastasis*	201 (80)	0	2 (1)
Malignant tumor, primary uncertain†	46 (18)	3 (8)	7 (4)
Malignant tumor, not melanoma metastasis	1 (<1)	17 (42)	26 (14)
No or insufficient evidence of malignant tumor	0	12 (30)	95 (53)
Insufficient evidence to classify	1 (<1)	7 (18)	41 (23)
Not reviewed‡	2 (<1)	1 (2)	9 (5)

Abbreviation: COMS, Collaborative Ocular Melanoma Study.

*Indicates histopathologically confirmed melanoma metastasis was present at the time of death.

†Indicates melanoma metastasis was suspected but was not confirmed histopathologically.

‡Indicates deaths discovered after the final meeting of the Mortality Coding Committee.